

Isentropic Efficiency of Adiabatic Fan and Compressor

$$\cancel{(\dot{Q}_{\text{IN}} - \dot{Q}_{\text{OUT}})_{\text{Adiabatic}}} + (\dot{W}_{\text{IN}} - \dot{W}_{\text{OUT}}) = \dot{m}(h_{0,\text{OUT}} - h_{0,\text{IN}})$$

$$\dot{W}_{\text{IN}} = \dot{W}_{\text{shaft}} = \int \rho \dot{w}_{\text{shaft}} dV$$

$$\rho \dot{w}_{\text{shaft}} = \rho \vec{u} \cdot \nabla w_{\text{shaft}} = \rho \vec{u} \cdot \vec{f}_{\text{shaft}} \quad (\text{Per unit volume})$$

This because viscous work opposes the “direction” of work which is transferring outwards

$$\dot{W}_{\text{OUT}} = -\dot{W}_{\text{shear}}; \quad \dot{W}_{\text{shear}} = \int (\tau \vec{u}) \cdot \hat{n} ds \quad \begin{array}{l} \text{where } \hat{n} \text{ points } \mathbf{OUT-} \\ \mathbf{WARDS} \text{ of control sur-} \\ \text{face.} \end{array}$$

$$\therefore \dot{W}_{\text{shear}} = \int \nabla \cdot (\tau \vec{u}) dV$$

$$\rightarrow \dot{W}_{\text{shaft}} + \dot{W}_{\text{shear}} = \dot{m}(h_{0,\text{OUT}} - h_{0,\text{IN}})$$

$$\dot{W}_{\text{shaft}} = \dot{m}(h_{0,\text{OUT},s} - h_{0,\text{IN}})$$

$$(1) \quad \eta_s = \frac{h_{0,\text{OUT},s} - h_{0,\text{IN}}}{h_{0,\text{OUT}} - h_{0,\text{IN}}} = \frac{\dot{m}(h_{0,\text{OUT},s} - h_{0,\text{IN}})}{\dot{m}(h_{0,\text{OUT}} - h_{0,\text{IN}})} = \frac{\dot{W}_{\text{shaft}}}{\dot{W}_{\text{shaft}} + \dot{W}_{\text{shear}}}$$

$$\eta_s = \frac{1}{1 + \frac{\dot{W}_{\text{shear}}}{\dot{W}_{\text{shaft}}}} = \frac{1}{1 + \frac{\int \nabla \cdot (\tau \vec{u}) dV}{\int \rho \dot{w}_{\text{shaft}} dV}} = \frac{1}{1 + \frac{\int \nabla \cdot (\tau \vec{u}) dV}{\int \rho \vec{u} \cdot \vec{f}_{\text{shaft}} dV}}$$

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$$\eta_s = \frac{1}{1 + \frac{\int \nabla \cdot (\tau \vec{u}) dV}{\dot{W}_{\text{shaft}}}}$$

$$\eta_s = \frac{1}{1 + \frac{\int \nabla \cdot (\tau \vec{u}) dV}{\int \rho \dot{w}_{\text{shaft}} dV}}$$

$$\Rightarrow \eta_s(x, y) = \frac{1}{1 + \frac{\nabla \cdot (\tau \vec{u})}{\rho \dot{w}_{\text{shaft}}}}$$

$$\eta_s = \frac{1}{1 + \frac{\int \nabla \cdot (\tau \vec{u}) dV}{\int \rho \vec{u} \cdot \vec{f}_{\text{shaft}} dV}}$$

$$\Rightarrow \eta_s(x, y) = \frac{1}{1 + \frac{\nabla \cdot (\tau \vec{u})}{\rho \vec{u} \cdot \vec{f}_{\text{shaft}}}}$$

Where $\dot{w}_{\text{shaft}}$ and $\vec{f}_{\text{shaft}}$ and $\dot{W}_{\text{shaft}}$

can be treated as constant over a determined control volume

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$$\eta_s = \frac{\dot{W}_{\text{shaft}}}{\dot{W}_{\text{shaft}} + \dot{W}_{\text{shear}}} \quad \text{from equation (1)}$$

$$\eta_s = \frac{\int \rho \vec{u} \cdot \vec{f}_{\text{shaft}} dV}{\int \rho \vec{u} \cdot \vec{f}_{\text{shaft}} dV + \int \nabla \cdot (\tau \vec{u}) dV}$$

$$\nabla \cdot (\tau \vec{u}) = u \cdot (\nabla \cdot \tau) + \tau : \nabla \vec{u}$$

Momentum N-S Equation

$$\rho(\vec{u} \cdot \nabla) \vec{u} = -\nabla P + \nabla \cdot \tau + \rho \vec{f}_{\text{shaft}}$$

$$\textcircled{1} \quad \rho(\vec{u} \cdot \nabla) \vec{u} + \nabla P = \nabla \cdot \tau + \rho \vec{f}_{\text{shaft}}$$

Now, P_0 is pressure at a spatial point

where fluid has come to rest

adiabatically and reversibly

$$(\delta q_{\text{IN}} - \delta q_{\text{OUT}})_{\text{Adiabatic}} + (dw_{\text{IN}} - \underbrace{dw_{\text{OUT}}}_{\text{no irreversible viscous work}}) = dh_0$$

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$$dw_{\text{in}} = dw_{\text{shaft}}$$

$$\therefore dw_{\text{shaft}} = dh_0$$

$$dw_{\text{shaft}} = dh + u du; \quad dh_0 = d\left(h + \frac{1}{2}u^2\right)$$

$$Tds = dh - \frac{dP}{\rho} = 0 \rightarrow dh = \frac{dP}{\rho}$$

0, isentropic

$$\rightarrow dw_{\text{shaft}} = \frac{dP}{\rho} + u du$$

$$\frac{dw_{\text{shaft}}}{d(x, y)} = \frac{1}{\rho} \frac{dP}{d(x, y)} + \frac{u du}{d(x, y)}$$

$$\nabla w_{\text{shaft}} = \frac{\nabla P}{\rho} + (\vec{u} \cdot \nabla) \vec{u}$$

$$\rightarrow \vec{f}_{\text{shaft}} = \frac{\nabla P}{\rho} + (\vec{u} \cdot \nabla) \vec{u}$$

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$$\vec{f}(x, y, \vec{u}, t) = \vec{f}, \text{ it is constant}$$

$$\therefore \frac{\nabla P}{\rho} + (\vec{u} \cdot \nabla) \vec{u} - \vec{f}_{\text{shaft}} = 0$$

for all (x, y) isentropic

$$\text{when } \vec{u} = \vec{0}, \quad P = P_0 \quad \text{and} \quad \rho = \rho_0$$

$$\therefore \frac{\nabla P_0}{\rho_0} + (\vec{0} \cdot \nabla) \vec{0} - \vec{f}_{\text{shaft}} = 0$$

$$\frac{\nabla P_0}{\rho_0} - \vec{f}_{\text{shaft}} = 0 = \frac{\nabla P}{\rho} + (\vec{u} \cdot \nabla) \vec{u} - \vec{f}_{\text{shaft}}$$

$$\therefore \frac{\nabla P_0}{\rho_0} = \vec{f}_{\text{shaft}} \quad \left(\textcircled{2} \frac{\nabla P_0}{\rho_0} - \vec{f}_{\text{shaft}} = \frac{\nabla P}{\rho} + (\vec{u} \cdot \nabla) \vec{u} - \vec{f}_{\text{shaft}} \right)$$

$$\textcircled{2} \quad \frac{\nabla P_0}{\rho_0} = \frac{\nabla P}{\rho} + (\vec{u} \cdot \nabla) \vec{u}$$

$$\therefore \textcircled{2} \quad \frac{\rho}{\rho_0} \nabla P_0 = \nabla P + \rho (\vec{u} \cdot \nabla) \vec{u}$$